

Command line arguments

```
graph TD; INPUT([INPUT]) --> Compile[Compile time input]; INPUT --> Command[Command line input]; INPUT --> Runtime[Runtime input];
```

INPUT

Compile time input

Command line input

Runtime input



Compile time input

Merit:-

Faster execution.

Demerit :-

Re-compilation is required to modify the value.

Ex : `int x = 10;`





Run-time input

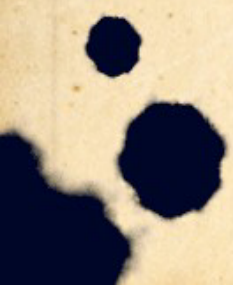
Merit:-

Re-compilation is not
Required to modify the value.

Demerit :-

Slower in execution.

Ex : `scanf("%d",&x);`





Command line input

Merit:-

Faster in execution & re-compilation is not Required to modify the value..

Demerit :-

Conversion functions are required to take a proper input for integers and floats..

Ex : `./a.out 10 18.9 vector`

Command line arguments (loadtime input)

Providing the arguments to a program at command prompt along with executable file is called as command line arguments.

These inputs we are giving to a program at the time of executable file loading into main memory. So we also call it as loadtime input.

Ex:- ./a.out 1234 22.7 A vector



These arguments will be received by:-
`Int main(int argc , char** argv).`

main() can be written in 3 ways

1. `int main()`
2. `int main(int argc, char *argv[ ]);`
3. `int main(int argc, char *argv[ ], char*env[ ]);`

But , 2 and 3 only follows command line arguments.